MAKHANA FARMING

Course - DS5003| Ergonomics in Product Design

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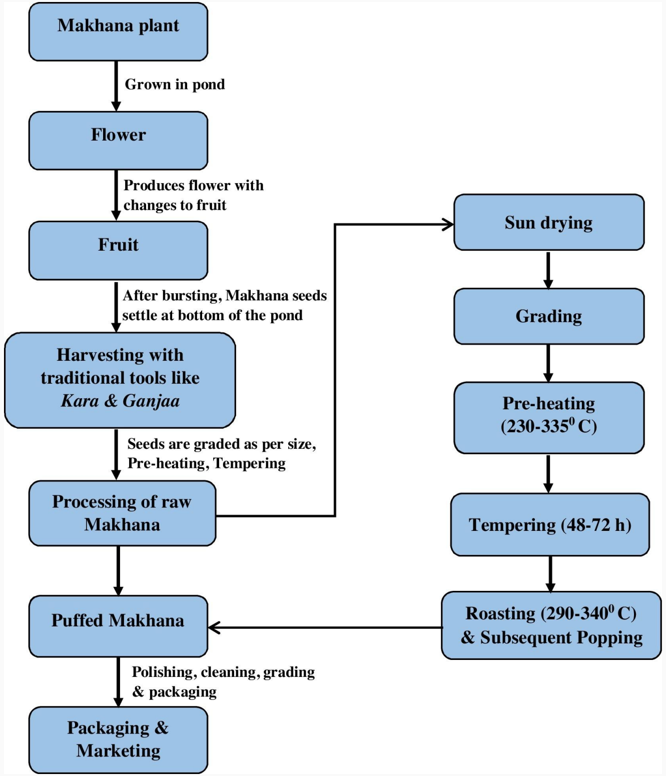
Introduction

Introduction

Makhana (Euryale ferox), also known as fox nut or gorgon nut, is an aquatic cash crop mostly cultivated in lowland ponds of Bihar, Orissa, Assam and West Bengal. In India, about 80% of total Makhana comes from Darbhanga, Madhubani, Purnia and Katihar district of Bihar and over 85% of global supply, earning it the nickname "Black Gold of Bihar." This superfood is rich in protein, fiber, and antioxidants, gaining rapid international popularity as a healthy snack alternative.

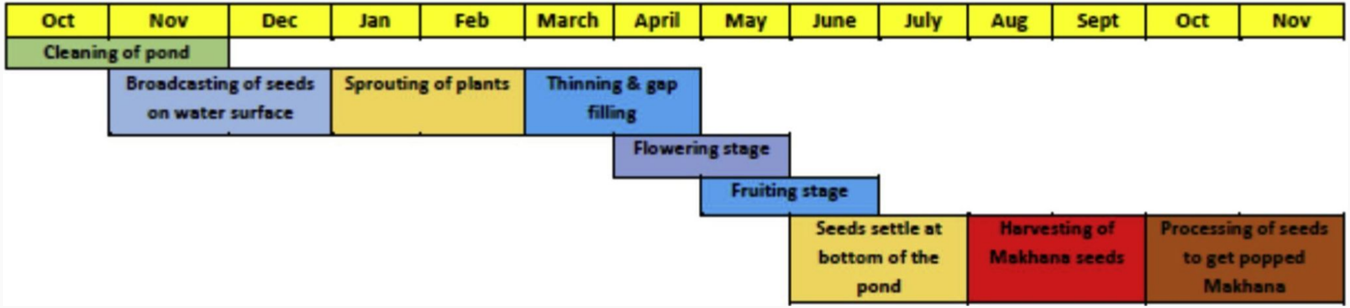
Cycle of Makhana production



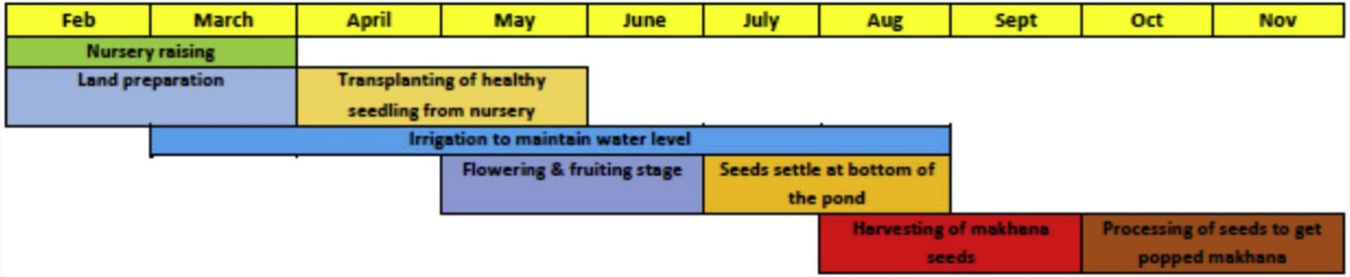
Makhana's Journey

Makhana Farming - Timeline

Makhana grown in ponds



Makhana grown in agricultural field



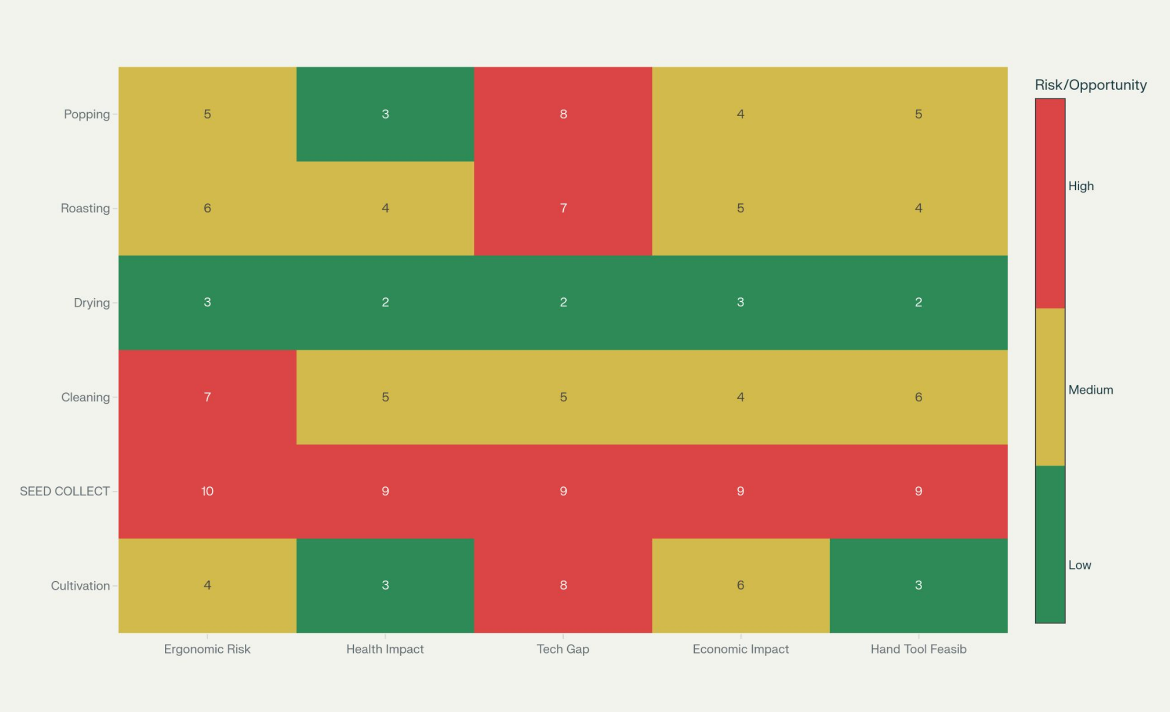
Problem Identification

Identifying Opportunity Gap

Why Seed Collection?

Criteria	Seed Collection	Other Stages
Ergonomic Risk	**Extreme:** 6-8 hrs submerged, prone positions, repeated sweeping	**Moderate:** Heat/repetitive motion
Health Impact	**Severe:** 78% lower back pain, 67% neck pain, 52% wrist pain, daily injuries	**Less severe:** Occupational stress only
Technology Gap	**Vast:** Unchanged bamboo tools vs. possible mechanical solutions	**Narrow:** Some mechanization attempted
Economic Impact	**Highest:** Bottleneck limiting entire chain productivity	Secondary impact
Hand Tool Feasibility	**High:** Affordable intermediate solution (Rs 1-5k) possible	**Low:** Requires major infrastructure changes
Workforce Crisis	**Critical:** Young workers abandoning job due to physical demands	Not applicable
Social Impact	**Highest:** Affects specialized Mallah/Sahni communities directly	Diverse workforce, less cultural significance

Production Stage Analysis Matrix



Expected Outcomes from Seed Collection Intervention

- ✓ 30-50% reduction in worker suffering & health risks
- ✓ 50-100% productivity improvement (4.0 → 6-8 kg/hr)
- ✓ Prevent industry collapse from labor shortage
- ✓ Create scalable, affordable solution for all farm sizes
- ✓ Cascade benefits through entire production chain
- ✓ Enhance global supply of superfood

Problem Overview

Current Reality:

- Workers spend 6-8 hours daily submerged in 4-6 feet murky water
- Prone underwater positions collecting seeds from pond beds
- Traditional bamboo tools unchanged for generations
- Sharp thorns covering plants cause constant injuries

Health Impact:

- 78% suffer chronic lower back pain
- 67% experience persistent neck pain
- 52% have chronic wrist/hand pain
- Daily thorn punctures leading to infections
- Working heart rate: 112.86 bpm ("heavy to very heavy" workload)

Economic Crisis:

- Productivity stagnant at 3.8-4.0 kg/hour
- 15.8% of seeds left uncollected (lost revenue)
- Workers earn only Rs 250-600 daily for extreme labor
- Younger generation abandoning occupation
- Labor shortages threatening Rs 3,000+ crore industry

Market Opportunities in India

India Market Size:

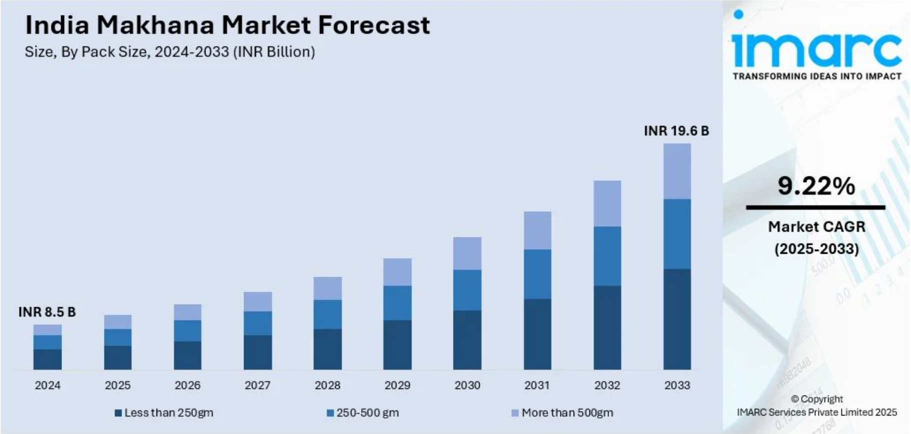
- **Current (2024):** INR 8.5 Billion
- **Projected (2033):** INR 19.6 Billion
- **CAGR:** 9.22%
- Bihar produces 80-90% of India's supply

Global Market:

- **Current (2023):** USD 44.4 Million
- **Projected (2030):** USD 97.5 Million
- **India:** World's largest producer & exporter (25,130+ shipments)

Bihar's Makhana Economy:

- **Current Value:** Rs 3,000 Crore
- **Expected (5 years):** Rs 6,000+ Crore
- **Government Support:** Makhana Board announced (Union Budget 2025)
- **Employment:** Hundreds of thousands directly/indirectly



Literature Study

Makhana-Specific Patents

1. Manually Operated Makhana Harvester with Optical Device

- **Source:** Central Agricultural University (2019)
- **Features:** Semi-mechanized with optical assistance for seed visibility
- **Status:** Research prototype

2. Mechanized Popping System (Patent 674/DEL/2013)

- **Source:** ICAR-CIPHET Ludhiana
- **Capacity:** 40 kg/hr
- **Focus:** Post-harvest processing, not collection

3. Bioactive Compound Patent (July 2025)

- **Source:** Bihar Agricultural University
- **Discovery:** N-(2-iodophenyl) methanesulfonamide in makhana
- **Significance:** First natural source, anti-microbial/anti-cancer potential

4. Makhana Cultivation Method Patent (Nov 2002)

- **Inventor:** Purshottam Bhagat
- **Focus:** Storage-stable edible food preparation

Mechanical Harvesters

Mud-Pump Mechanical Harvesters:

- **Productivity:** 12.6 kg/hr (3x manual)
- **Labor Reduction:** 15 → 3 workers per hectare
- **Seed Recovery:** +15.8% over manual
- **Cost:** Rs 1,00,000-3,00,000 (unaffordable for small farmers)
- **Limitation:** Requires power source, maintenance, technical expertise

Improved Diving Kits:

- Waterproof clothing, thorn-resistant gloves, diving masks
- **Result:** Productivity increase to 11.3 kg/hr
- **Limitation:** Still requires full submersion and awkward postures

Gap Analysis

What Exists:

- Expensive mechanical harvesters (Rs 1-3 lakh)
- Traditional bamboo tools (no improvements)
- Protective gear (reduces injury but not workload)

What's Missing:

- Affordable intermediate solution (Rs 500-5,000 range)
- Ergonomically designed hand tool reducing postural stress
- Protection integrated into tool design
- Mechanical advantage for seed dislodgement
- Built-in sieving during collection
- Reduced submersion time requirement

Existing Products/ Methods

Traditional Tools (Current Use)

Kaara (Bamboo Anchor Pole):

- **Length:** 2-3 meters
- **Function:** Anchoring reference point in pond
- **Limitations:** No collection assistance, requires full submersion

Ganjaa (Bamboo Sieve):

- Cone-shaped, 40-60 cm diameter
- **Function:** Separates seeds from mud by density
- **Limitations:** No ergonomic handles, causes hand fatigue, requires hundreds of vigorous shaking motions

Research Based Design Insights

Ergonomic Findings:

- Prolonged prone/bent postures cause 78% lower back pain rate
- Repetitive underwater sweeping motions = high musculoskeletal stress
- No mechanical advantage in current tools
- Zero protection from thorns and contaminated water

Anthropometric Considerations:

- North-Eastern Indian workers have smaller body dimensions
- Tools must fit local population anthropometry
- Grip strength, reach, and postural capabilities vary by region

Productivity Analysis:

- **Manual:** 4.0 kg/hr | **Improved Diving Kit:** 11.3 kg/hr | **Mechanical:** 12.6 kg/hr
- **Opportunity:** Hand tool can target 6-8 kg/hr (50-100% improvement)
- **Labor reduction potential:** 15 workers → 8-10 workers per hectare

Aim and Objectives

Design Brief

Our Mission

To design an affordable, ergonomically optimized hand tool that empowers makhana farmers and workers to collect seeds safely, comfortably, and productively while preserving traditional knowledge and ensuring industry sustainability.

Primary Goal:

Enable underwater seed collection with 50-100% productivity improvement while reducing worker health risks and physical suffering

Design Principles:

- Build upon existing harvesting knowledge
- Use locally available materials and repair methods
- Design with and for the Mallah/Sahni community
- Enhance both environmental and social sustainability
- Affordable for small-scale farmers

Specifica Targets:

- **Ergonomic:** 30-50% reduction in postural stress and submersion time
- **Safety:** 50% reduction in thorn-related injuries
- Productivity: Increase from 4.0 to 6-8 kg/hr per worker
- **Economic:** Tool cost under Rs 5,000, payback within 2 seasons
- **Usability:** Learnable in 1 day, maintainable locally
- **Durability:** 3-5 years lifespan in aquatic environment

Target Audiences

Makhana Harvesters (Direct Users)

- **Profile:** Men age 25-55 from Mallah/Sahni communities
- **Experience:** Traditional harvesting for 10-35 years
- Location: Mithilanchal region (Darbhanga, Madhubani, Samastipur)
- **Pain Points:** Chronic musculoskeletal pain, daily thorn **injuries, low earnings**
- **Needs:** Reduced physical strain, better safety, higher productivity

Small-Scale Farmers (Buyers/Employers)

- **Profile:** Cultivating 1-5 acres of makhana ponds
- **Challenge:** Labor shortage, rising wages, unaffordable mechanization
- **Budget:** Limited capital (max Rs 1,000-5,000 for tools)
- **Needs:** Affordable solution improving harvest efficiency

Methodology

Interview Questions

- How do you cultivate and manage your makhana crop throughout the year?
- How many harvests do you perform each year? In which months?
- How much seed is collected in first harvest? How much in second harvest?
- Can you describe your seed collection process during both harvests?
- What tools or equipment do you currently use for harvesting?
- What does a typical day look like for you during the makhana harvesting season?
- While working in the pond or field during harvesting, what postures or body movements do you use most frequently? Which of these causes the most pain?
- How many hours do you usually spend submerged in water or bent over while working each day?
- Which part of the seed collection process do you find the most difficult or tiring?
- Have you ever been injured by thorns? How often does this happen?
- What other health problems or injuries occur during seed collection (e.g., cuts, infections, muscle pain)?
- Where do you feel the most muscle pain in your body after the end of the day?
- What precautions or remedies do you take to prevent or treat such injuries?
- How safe do you feel while working in the pond or muddy water during seed collection? What do you consider the biggest risk?

Interview Questions

- Usually, how many laborers are involved in the harvesting process? How is the payment or wage decided?
- Do you experience any seed loss during harvesting? Do you have an estimate of how much is lost?
- If the seed collection process were easier or faster, how would it affect the yield or profit?
- Have you ever tried any new tool or method for seed collection? If yes, which one worked best or worst for you?
- If you were provided with a low-cost and efficient tool for seed collection, would you use it?
- What qualities would be most important for you in a new tool (e.g., safety, simplicity, speed, or cost)?
- In your opinion, what is the most essential thing to make the seed collection process easier or safer?
- If you could change one thing in the seed collection process, what would it be?
- What advice would you give to someone who is designing a new product or tool for makhana harvesting?

Telephonic Interview Recording

Interview 1

Interview 2

Interview 3

Observational Study

Insights

Interview Key Insights

Research Claim	Interview Verification	Additional Detail
25-30% seed loss	Confirmed by Multiple Farmers	Scattering during collection sweeping; small seeds escape
No viable machine for collection	Confirmed Explicitly	"Without hands, cannot harvest"
Entire body thorn injuries	Confirmed with Detail	Removal requires sickle blade scraping; heat application
6-8 hour submersion	Confirmed Explicitly	Workers hold breath during submersion
72-hour total processing	Confirmed Explicitly	Specific timeline: pond extraction → cleaning → drying → grading
65-70% material loss	Confirmed in Processing	30 kg raw → 8-10 kg popped; 73% loss specifically noted
Water depth criticality	New Insight	5-ft better for productivity; 3-ft creates extreme difficulty
Single annual harvest	Confirmed	One sowing (Oct-Nov), one harvest window (Jul-Nov)

Observation Insights

Research Claim	Visual Evidence	Status
78% lower back pain	All harvesting images show bent postures	✓ CONFIRMED
Daily thorn injuries	Workers handling thorny plants without gloves	✓ CONFIRMED
6-8 hours daily submersion	Large workforce spread across pond suggests duration	✓ CONFIRMED
Murky water conditions	Brown, opaque water in all pond images	✓ CONFIRMED
Pre-industrial tools	No modern equipment visible in harvesting	✓ CONFIRMED
62% women in processing	Female worker visible in sorting operation	✓ CONFIRMED
Technology gap	Modern storage vs. manual harvesting	✓ CONFIRMED
Community-based labor	Groups working together	✓ CONFIRMED

Empathy Mapping

SAYS

- "My hands are covered in thorn marks and my whole body hurts from this work"
- "The pain gets worse each season, and I apply traditional medicines every night to treat wounds"
- "These bamboo tools haven't changed in generations - without hands, we cannot harvest"
- "Young people won't do this work anymore; why would they when they can earn more in cities?"
- "There must be a better way to do this - if someone designed something for this job, I'd use it immediately"

THINKS

- "This is my family's hereditary occupation for 3 generations, but the suffering and low earnings make me wonder if my children will do this"
- "My back pain, neck pain, and constant thorn injuries are just accepted as 'normal' for this work, but I know it shouldn't have to be this way"
- "I'm earning only 12-25% of the Rs 1,200-2,000/kg retail value of makhana while farmers get rich selling my work"
- "Machines exist but cost Rs 1-3 lakh - we can't afford them, even though mechanization could do this job in 36 hours instead of 360 hours"
- "I don't want to leave farming, but without better tools and conditions, this occupation will disappear completely"

Empathy Mapping

DOES

- Submerges in murky pond water 6-8 hours daily in bent-over, prone positions to manually sweep mud and collect seeds by touch alone
- Applies thorn removal techniques nightly (sickle blade scraping, heat application) and uses traditional medicinal oils to treat daily puncture wounds
- Uses bamboo sieves to separate seeds from mud through vigorous shaking, repeating this cycle 50-100 times per day across the pond
- Works in groups for safety while coordinating with other harvesters, negotiating for deeper water depths (5 feet preferred) to reduce physical strain
- Teaches younger family members the traditional techniques despite knowing they will likely abandon the work for urban employment

FEELS

- 78% lower back pain, 67% neck pain, 52% wrist pain, daily thorn injuries causing constant suffering while earning only Rs 250-600/day
- Resentment that his destroyed body earns 12-25% of product value (Rs 1,200-2,000/kg retail), with no solution despite decades of trying
- Skilled at difficult ancestral work but ashamed young people view this occupation as low-status and undesirable
- Accepts "this is just how it is" but desperately open to any innovation that could reduce suffering while preserving his occupation
- Fears occupation will disappear entirely as younger generation abandons it, threatening his livelihood and community identity

User Persona

Rajesh

Makhana Seed Harvester

38 years old | 22 years experience | Mallah Community

Demographics

- Darbhanga District, Mithilanchal Region, Bihar
- Primary school (5th standard)
- Married, 2 children; wife works in cleaning
- ₹50,000-70,000/year (seasonal work, 90-120 days)

Work Context

- 6-8 hours/day during harvest (Sep-Jan)
- ₹250-600/day (₹30-60/kg harvested)
- 3.8-4.0 kg/hour productivity
- Traditional bamboo tools: Kara & Ganjaa
- Works with 10-30 harvesters per pond

Values & Motivations

Identity: Proud of ancestral knowledge, but ashamed of low status

Values: Family tradition, community ties, occupational pride

Aspirations: Better tools while preserving occupation; educated children staying in farming

Pain Points

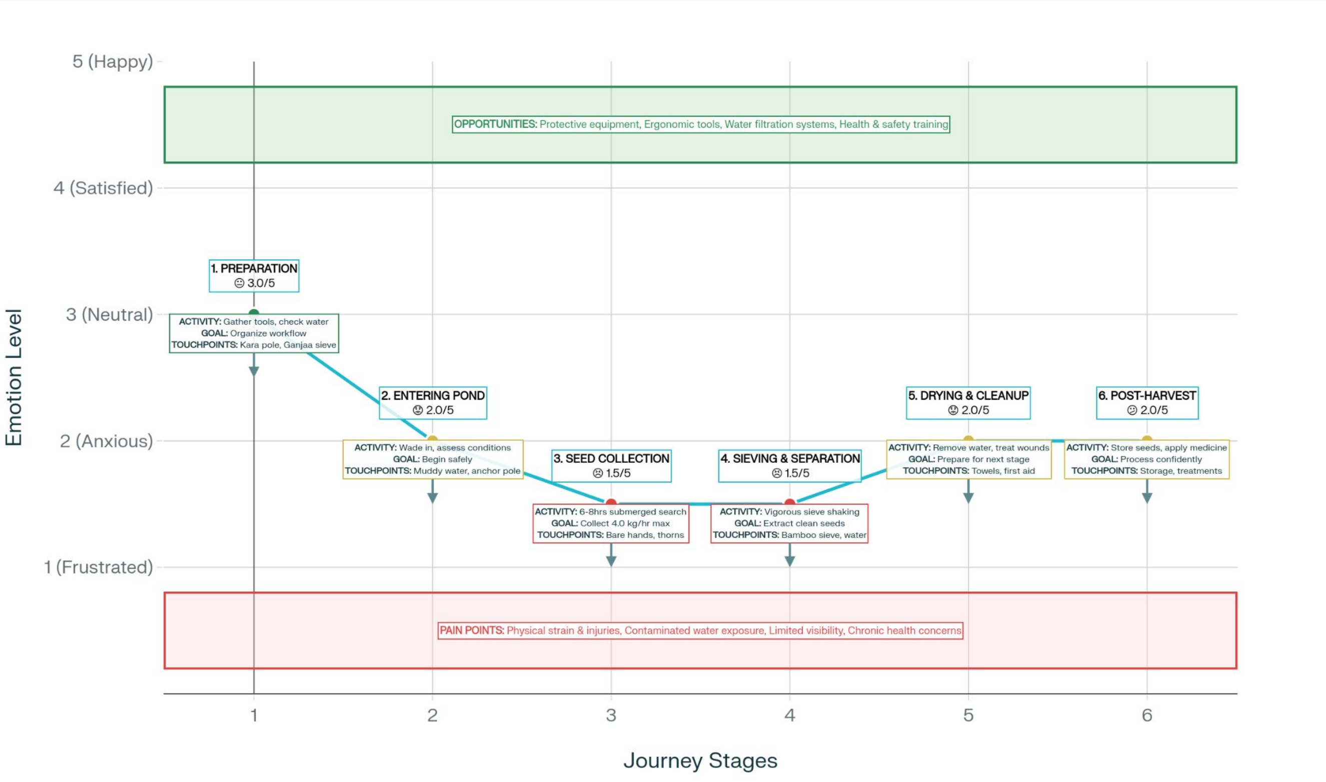
- Chronic pain: 78% lower back, 67% neck, 52% wrist
- Economic exploitation by intermediaries
- Workforce exodus threatening occupation
- Fear of permanent disability
- 25% seeds remain uncollected with current tools

Technology Comfort

Low digital literacy | Prefers hands-on demonstration | Values peer recommendations | Open to practical innovation

"I'm proud of what I do, but my body hurts. I want my children educated, but I hope they can stay in this work with better tools."

User Journey Mapping



Problem Statement

Workers in Bihar's makhana industry spend 6-8 hours daily submerged in murky pond water in extreme bent-over positions, using unchanged bamboo tools to manually collect seeds from muddy pond beds through tactile search alone. This pre-industrial harvesting method—the critical bottleneck in a modern high-value supply chain—traps workers in a cycle of severe physical suffering (78% lower back pain, 67% neck pain, 52% wrist pain, daily thorn injuries from thorny plants) while achieving stagnant productivity of just 3.8-4.0 kg/hour with 25% of seeds remaining uncollected.

Design Opportunity

Create an affordable (under Rs 5,000), ergonomically optimized hand tool that enables makhana seed collection with 30-50% reduced physical strain, 50% fewer thorn injuries, and 50-100% increased productivity (6-8 kg/hour) while remaining compatible with traditional harvesting knowledge and accessible to small-scale farmers—preserving occupational continuity while eliminating unnecessary suffering.

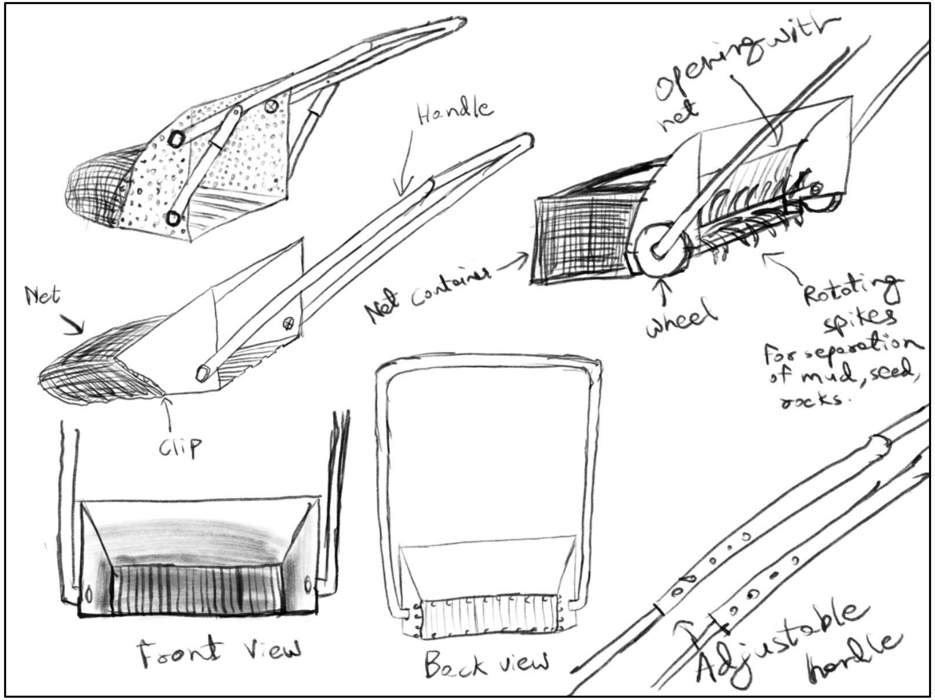
User Needs

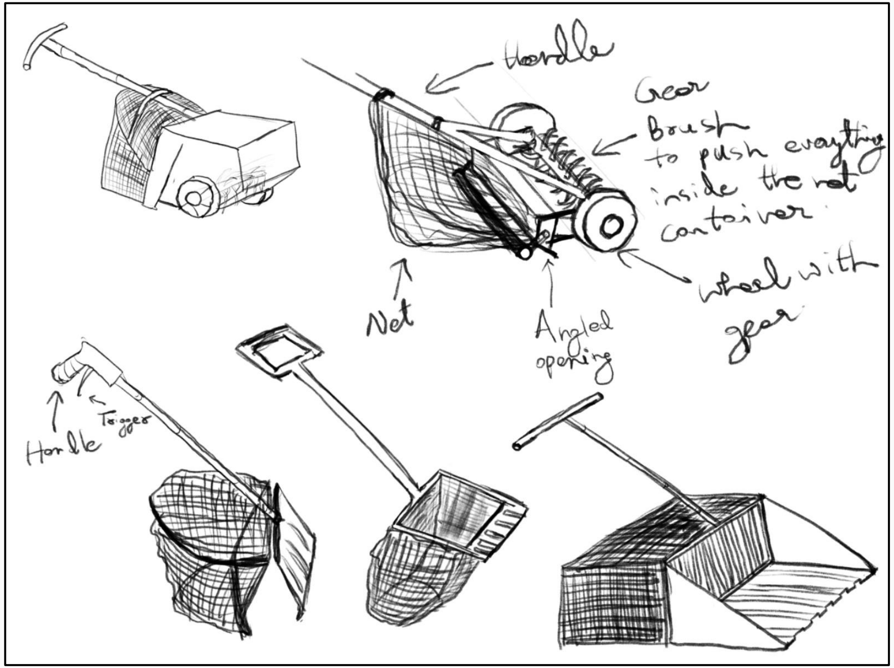
LEVEL	NEEDS	CURRENT STATE	DESIRED STATE
PRIMARY	Reduce pain and musculoskeletal strain	78% lower back pain, 67% neck pain, 52% wrist pain	30-50% pain reduction
	Prevent thorn injuries & infections	Nearly 100% daily puncture wounds	50% fewer injuries; zero infections
	Increase productivity & earnings	3.8-4.0 kg/hr; Rs 250-600/day	6-8 kg/hr; Rs 400-800/day
	Maintain occupational dignity	Shame, stigma, knowledge disappearing	Respected occupation; intergenerational continuity
SECONDARY	Reduce submersion time & postural stress	6-8 hrs daily underwater; extreme bending	4-6 hrs; upright working positions
	Improve seed recovery efficiency	25% seeds lost; low collection rates	Recover 5-10% additional seeds

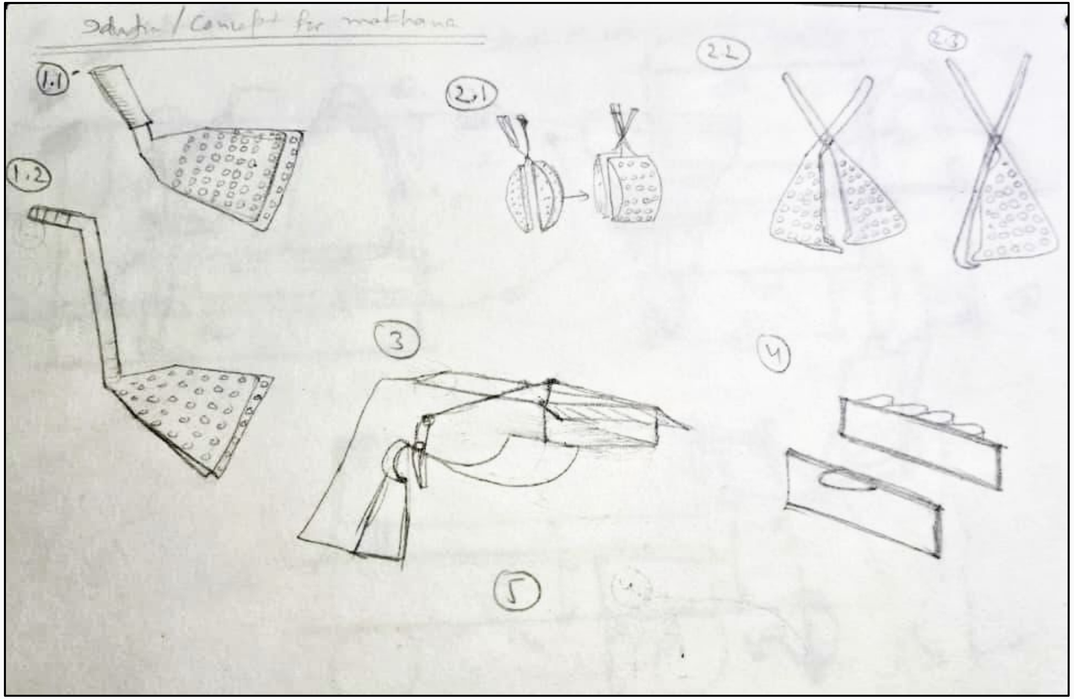
User Needs

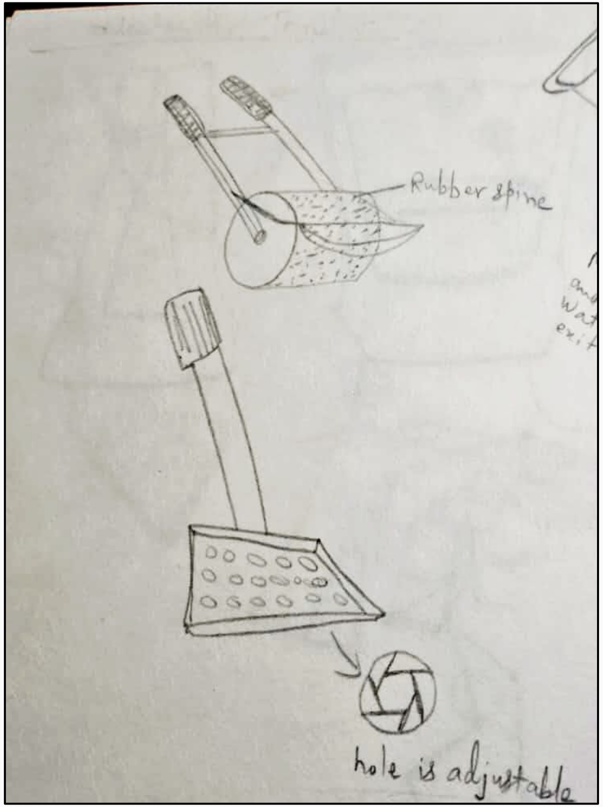
	Enable group-based harvesting	Workers interfere with each other	Multi-worker coordination without conflict
	Access affordable technology	Unaffordable machines (Rs 1-3 lakh)	Tool under Rs 5,000; 2-season payback
TERTIARY	Receive formal recognition & training	Zero occupational support systems	Industry recognition, mentorship programs
	Preserve cultural identity in innovation	Technology threatens traditional knowledge	Innovation builds on tradition
	Access health & safety support	No insurance, safety provisions, medical aid	Occupational protections, affordable health services
	Build collective bargaining power	Individual workers powerless	Cooperative organization for negotiation
	Achieve fair economic compensation	12-25% value capture (Rs 250-600 vs Rs 1,200-2,000/kg retail)	40-50% value capture; transparent pricing

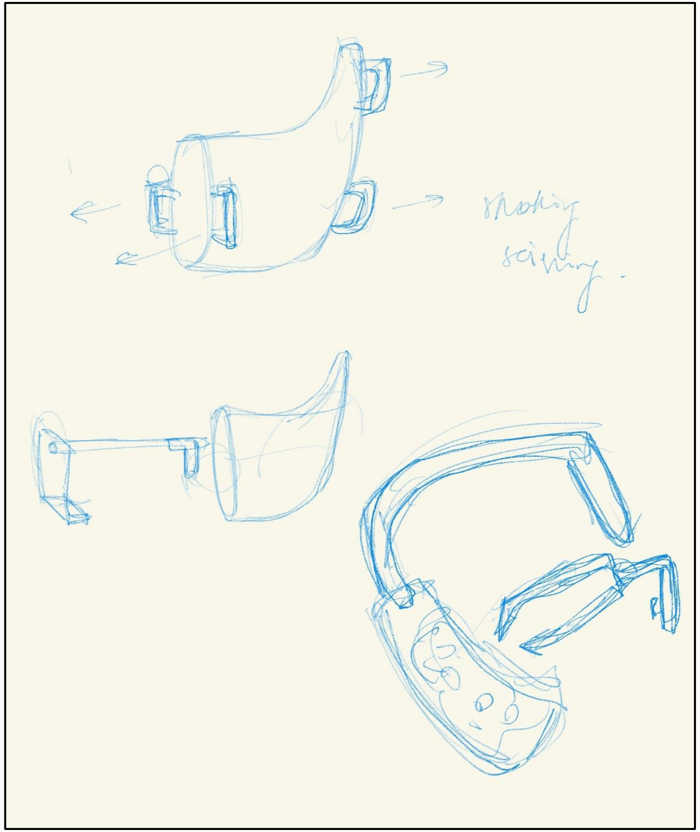
Ideation

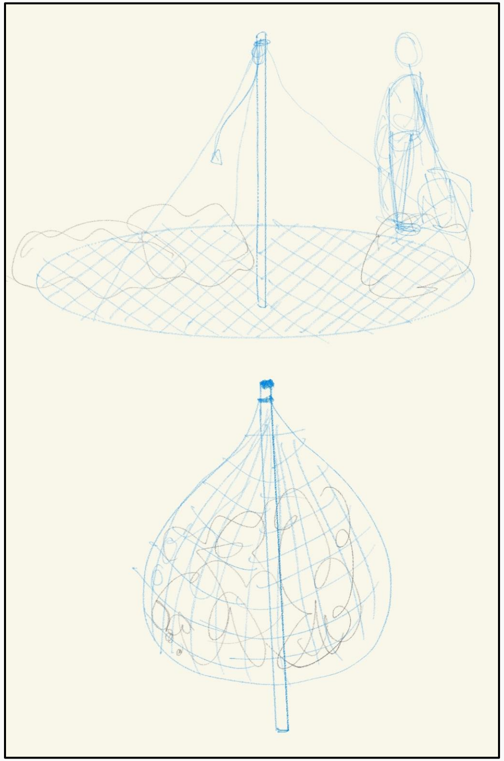

Handle
Opening with net
Net
Net container
Wheel
Rotating spikes
For separation of mud, seed, rocks.
Clip
Front view
Back view
Adjustable handle

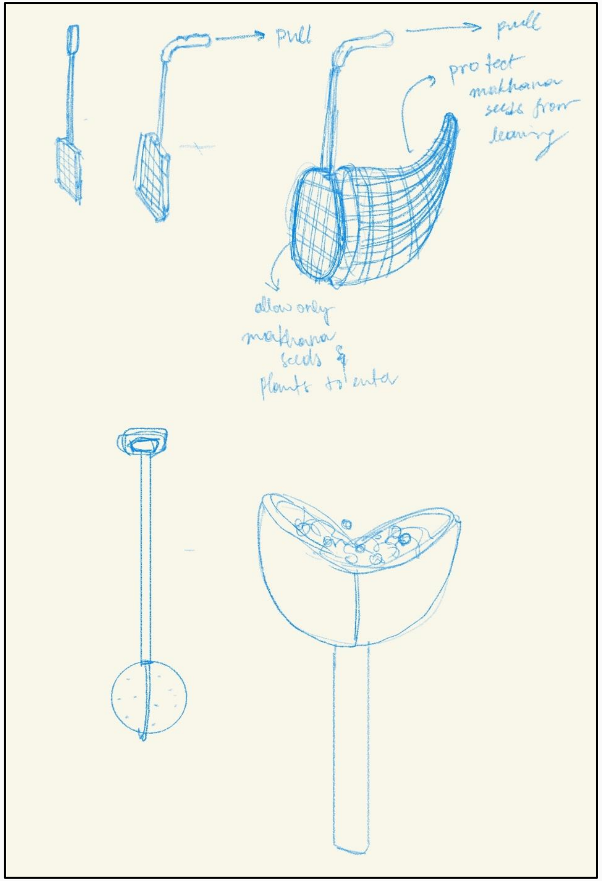

Handle
Gear
Brush to push everything inside the net container
Net
Angled opening
wheel with gear
Handle
Trigger

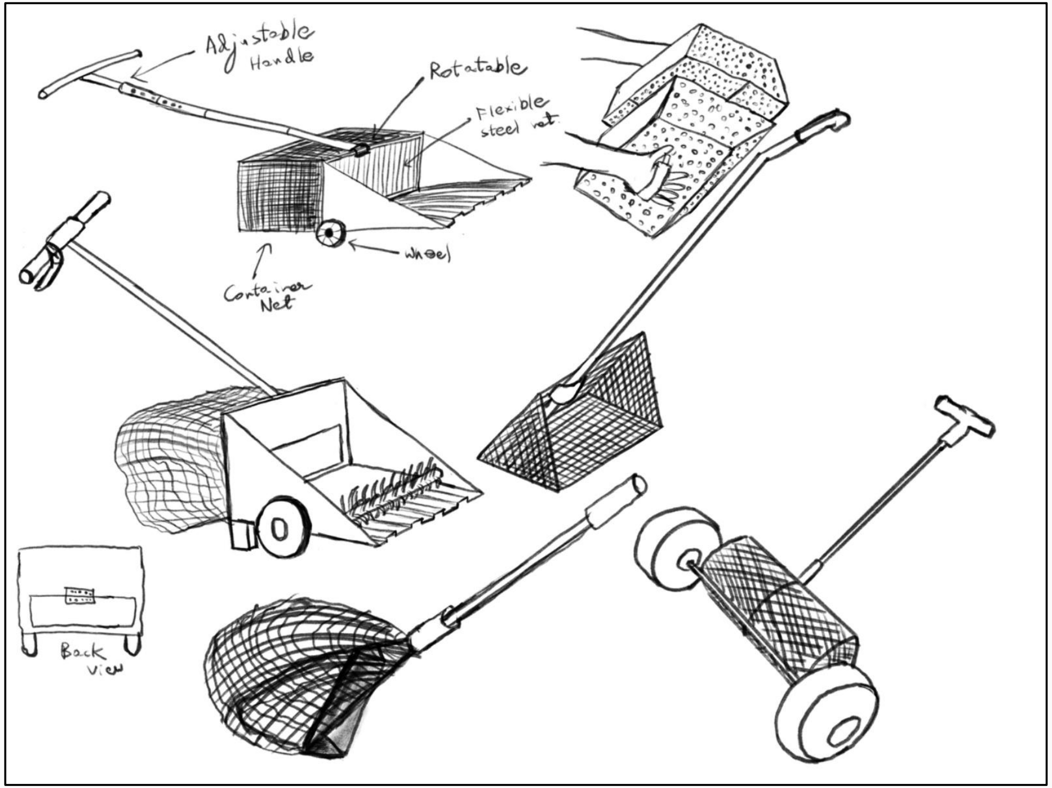

1.1
1.2
2.1
2.2
2.3
3
4
5

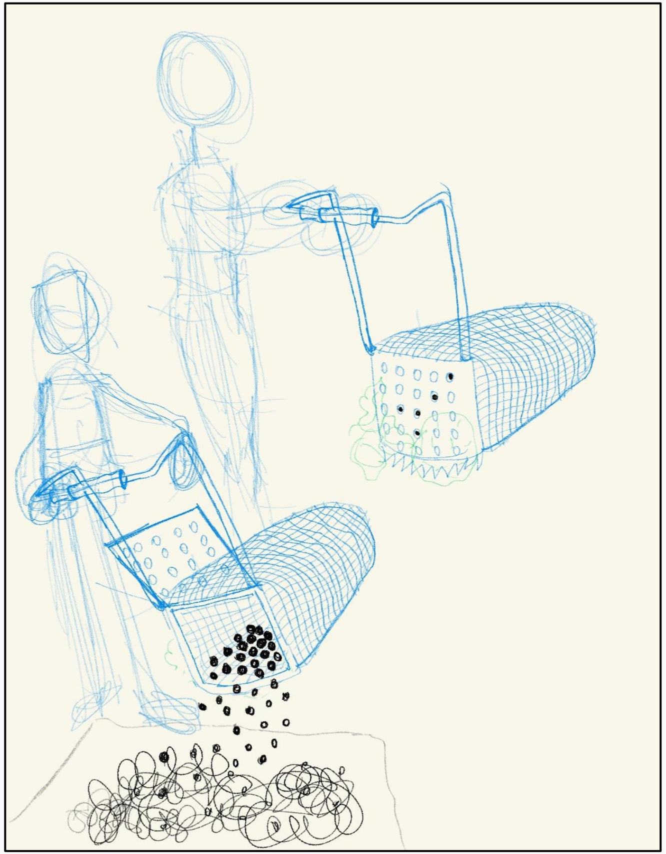

Rubber spine
hole is adjustable

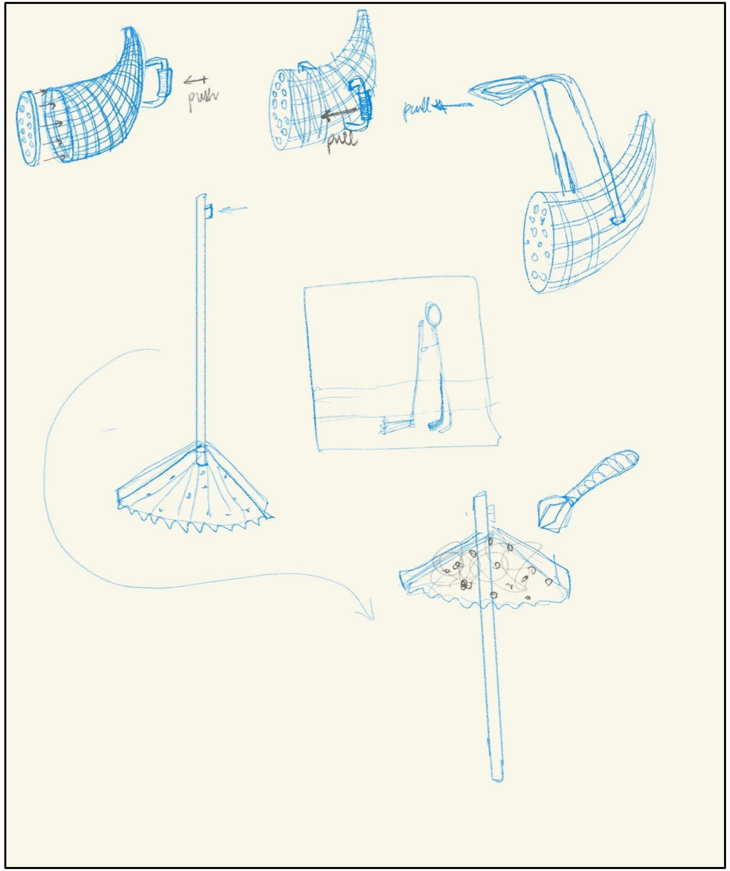

shaking
scissing

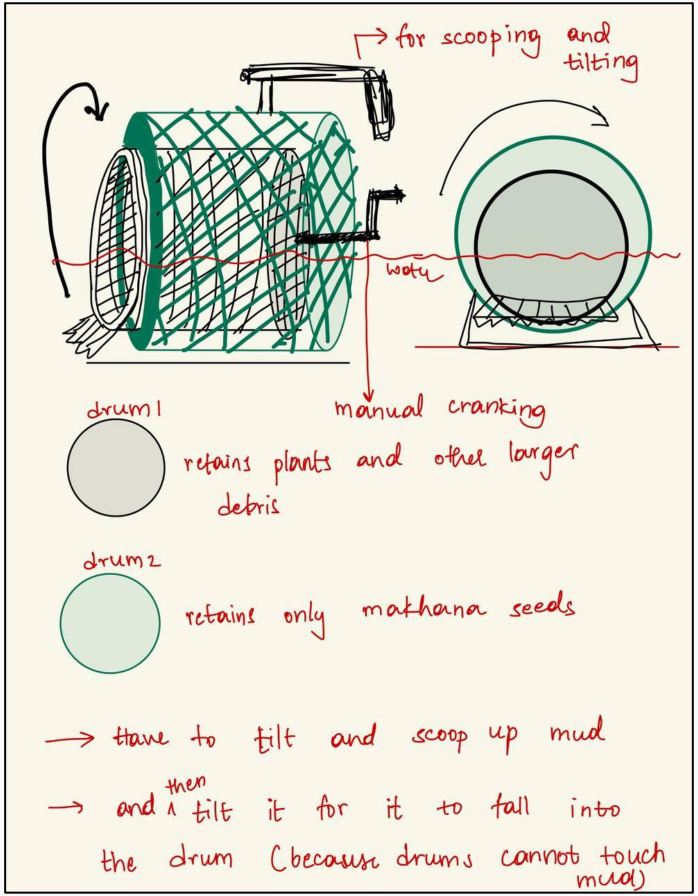


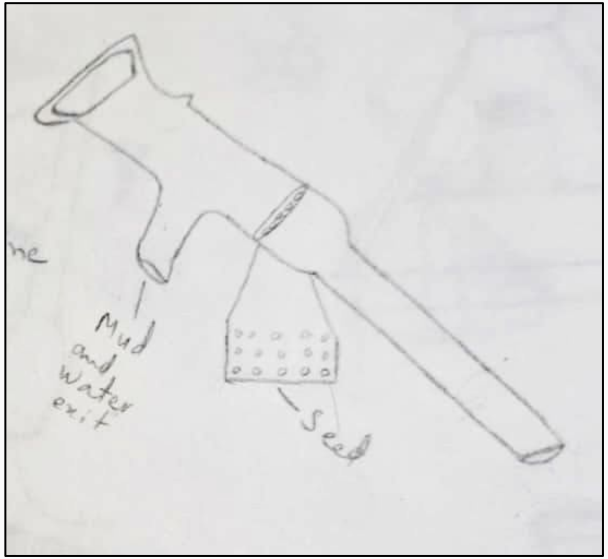

pull
pull
protect makhana seeds from leaving
allow only makhana seeds & plants to enter

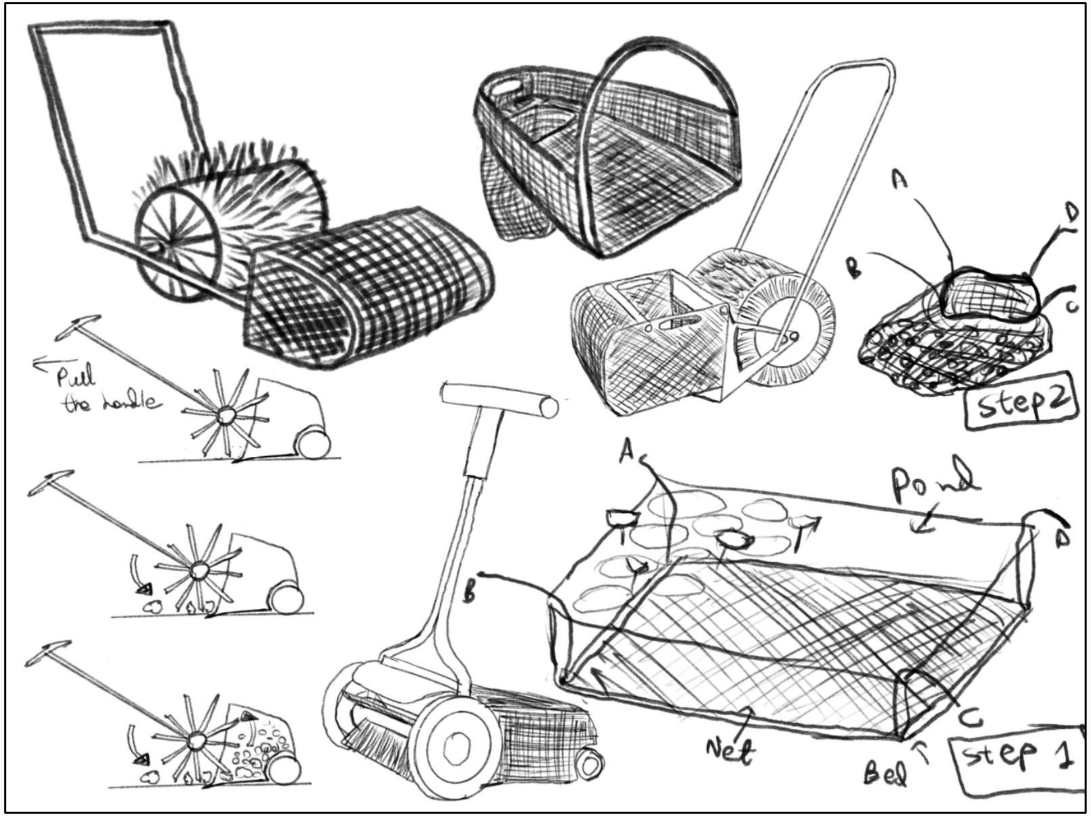

Adjustable Handle
Rotatable
Flexible steel net
wheel
Container Net
Back view




push
pull
pull


for scooping and tilting
water
manual cranking
drum1
retains plants and other larger debris
drum2
retains only makhana seeds
→ Have to tilt and scoop up mud
→ and then tilt it for it to fall into the drum (because drums cannot touch mud)

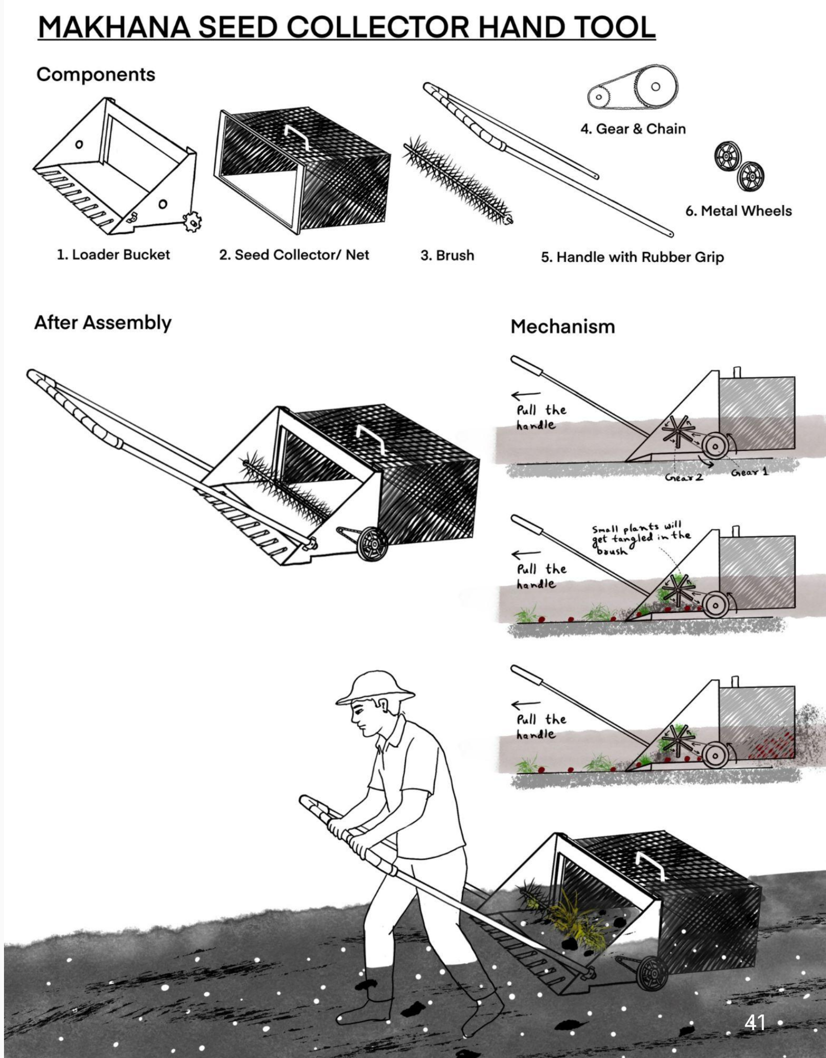

ne
Mud and Water exit
Seed


Pull the handle
A
B
C
D
Step2
Pond
Net
Bel
Step 1

Concept Finalization

Reasons for choosing concept

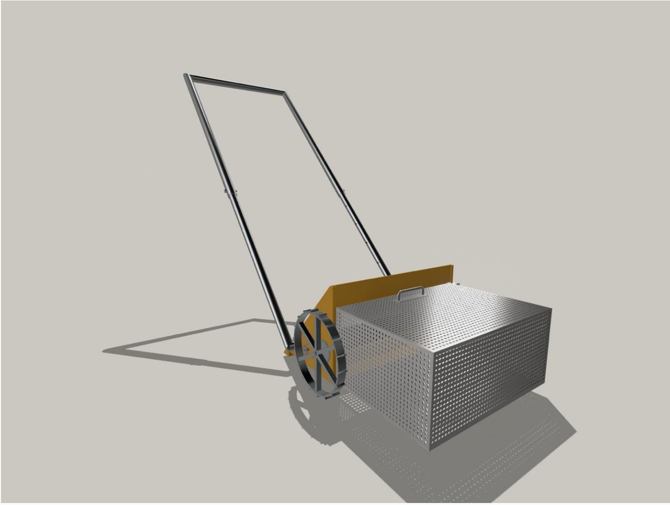
- Ease of pulling the product
- Brush that traps plants and debris
- Integration of both collection and first stage of cleaning
- Easy to lift and unload the basket
- Increased efficiency in collection of seeds



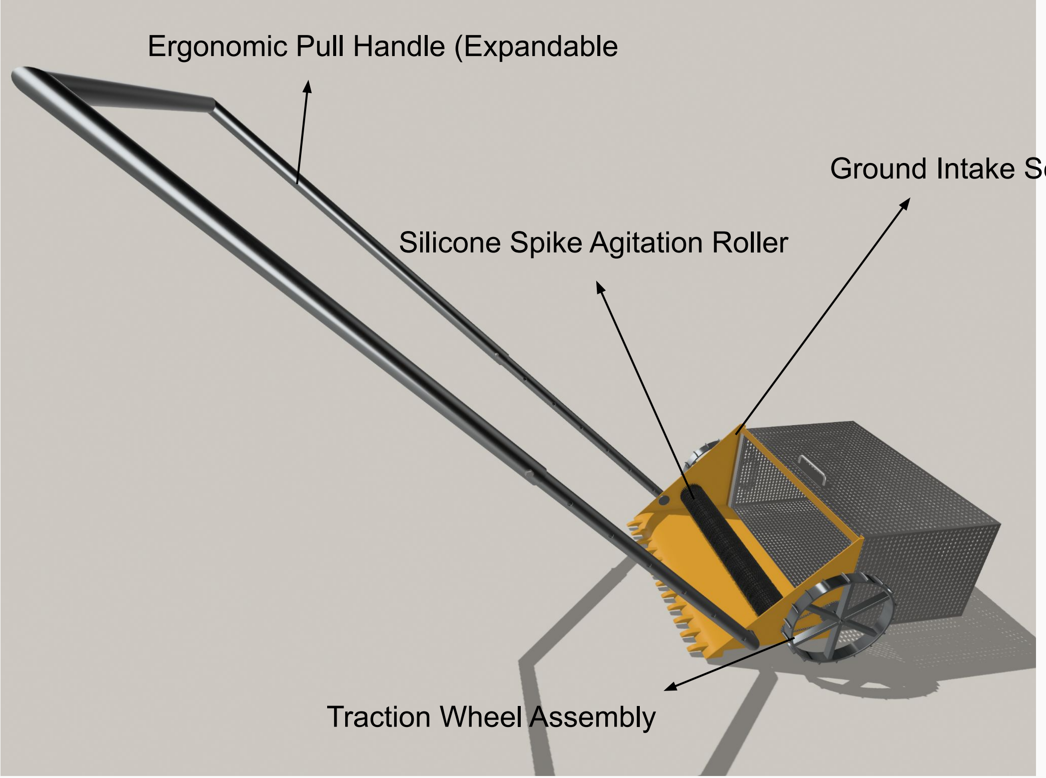
Technical aspects of product

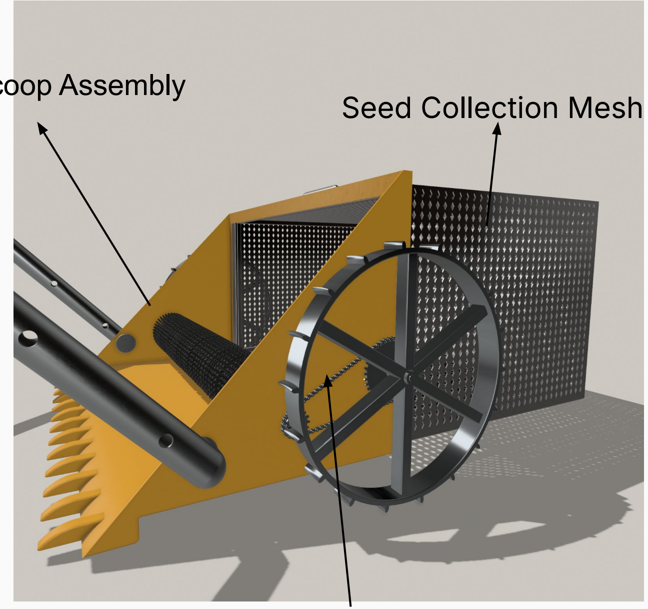
Materials

Part	Material
Loader bucket	Stainless steel
Seed collector basket	Stainless steel / Bamboo
Brush	Silicone
Handle	Stainless steel
Handle grip	Rubber
Wheels	Stainless steel / Iron

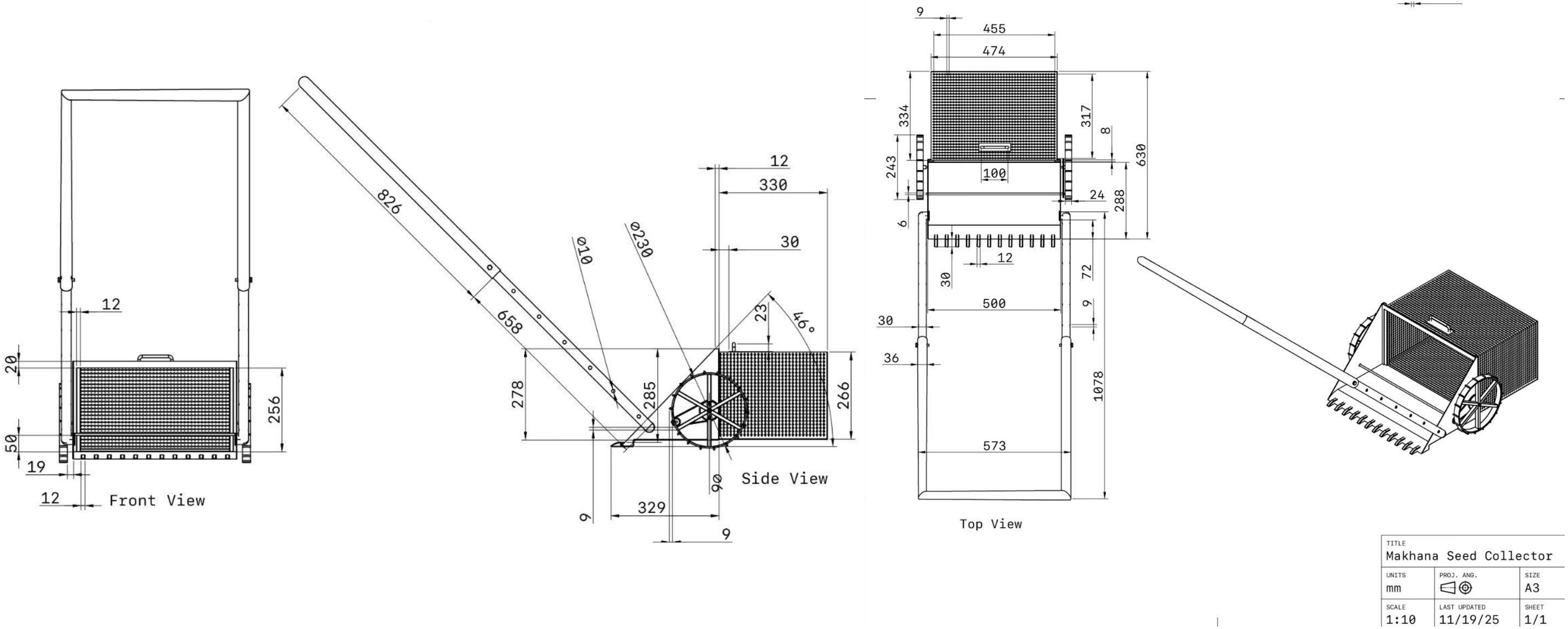


Product Specification

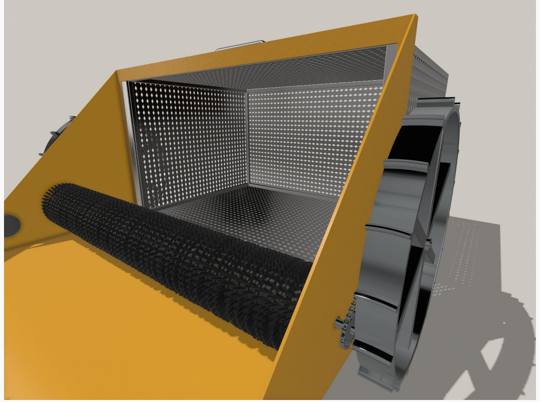


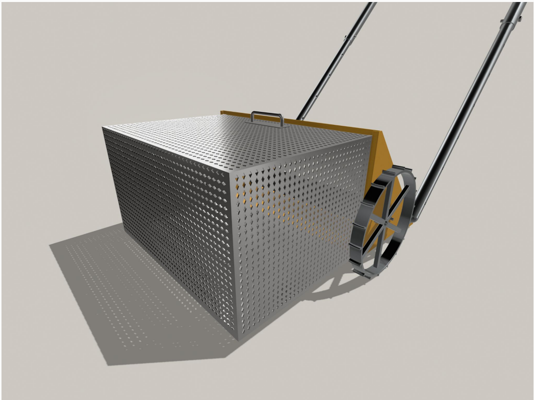


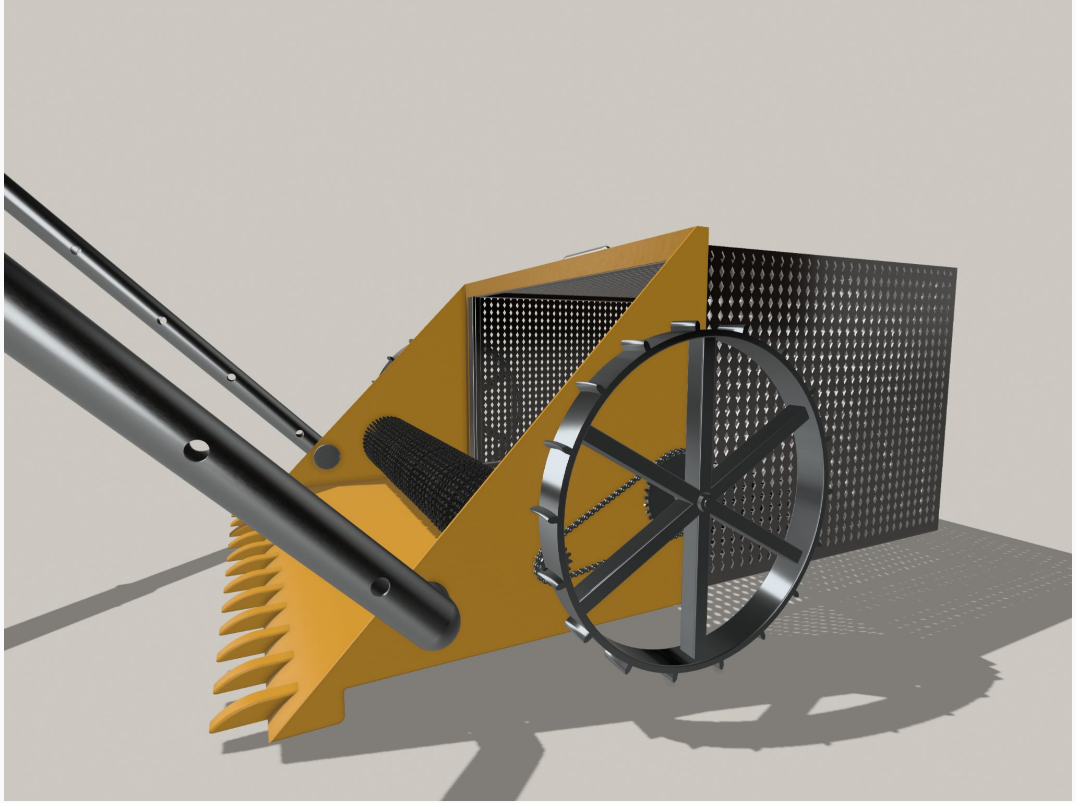
Dimensions


Front View
Side View
Top View
TITLE
Makhana Seed Collector
UNITS
mm
PROJ. ANG.
SIZE
A3
SCALE
1:10
LAST UPDATED
11/19/25
SHEET
1/1

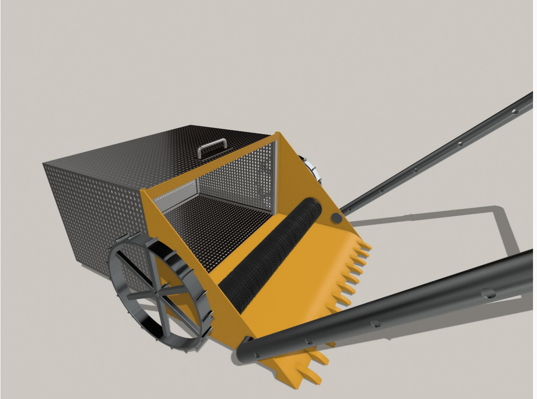
Renders

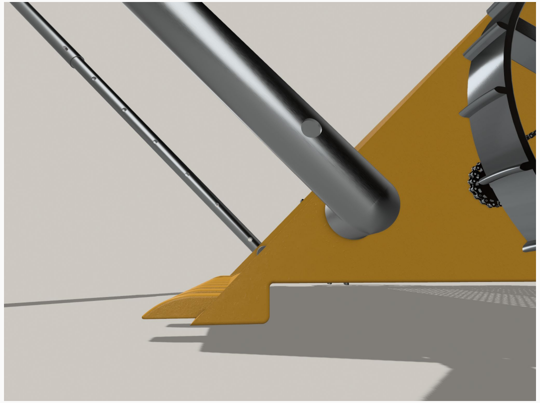


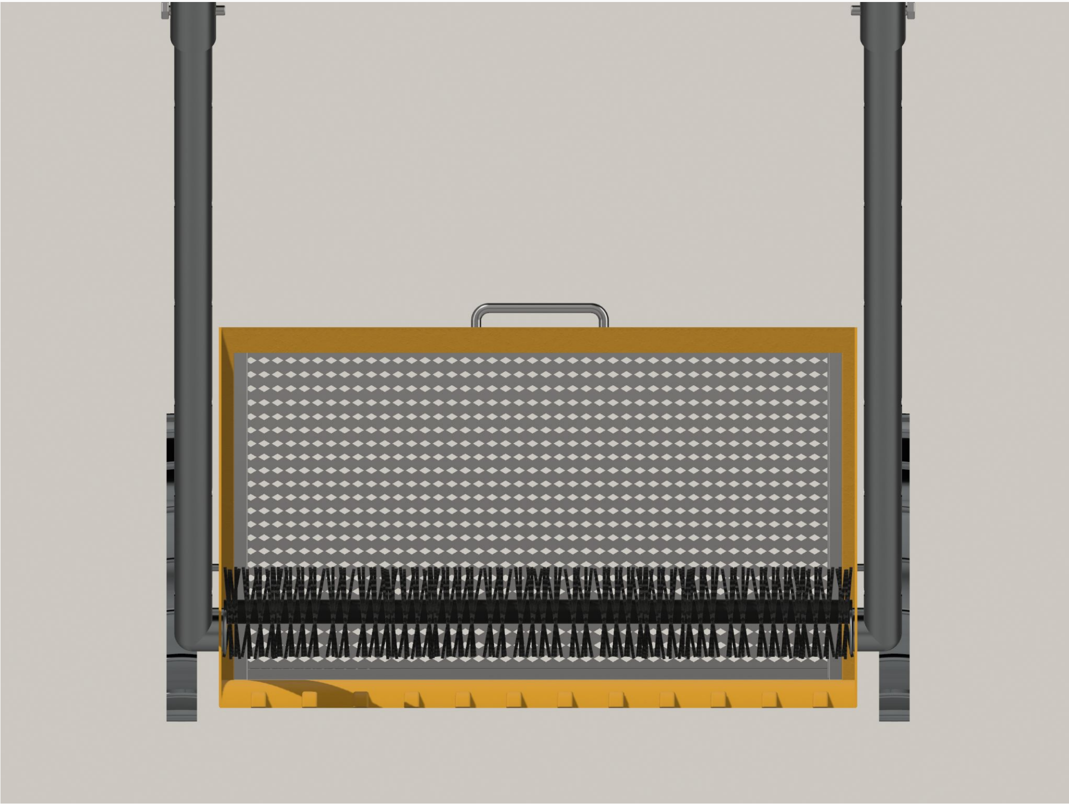




Renders







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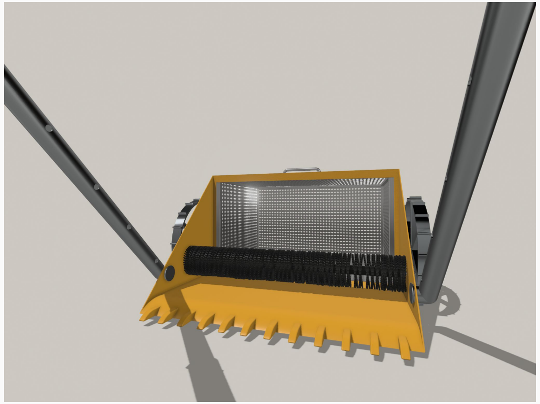

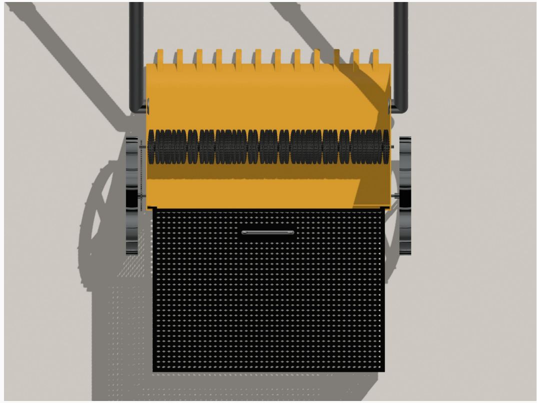

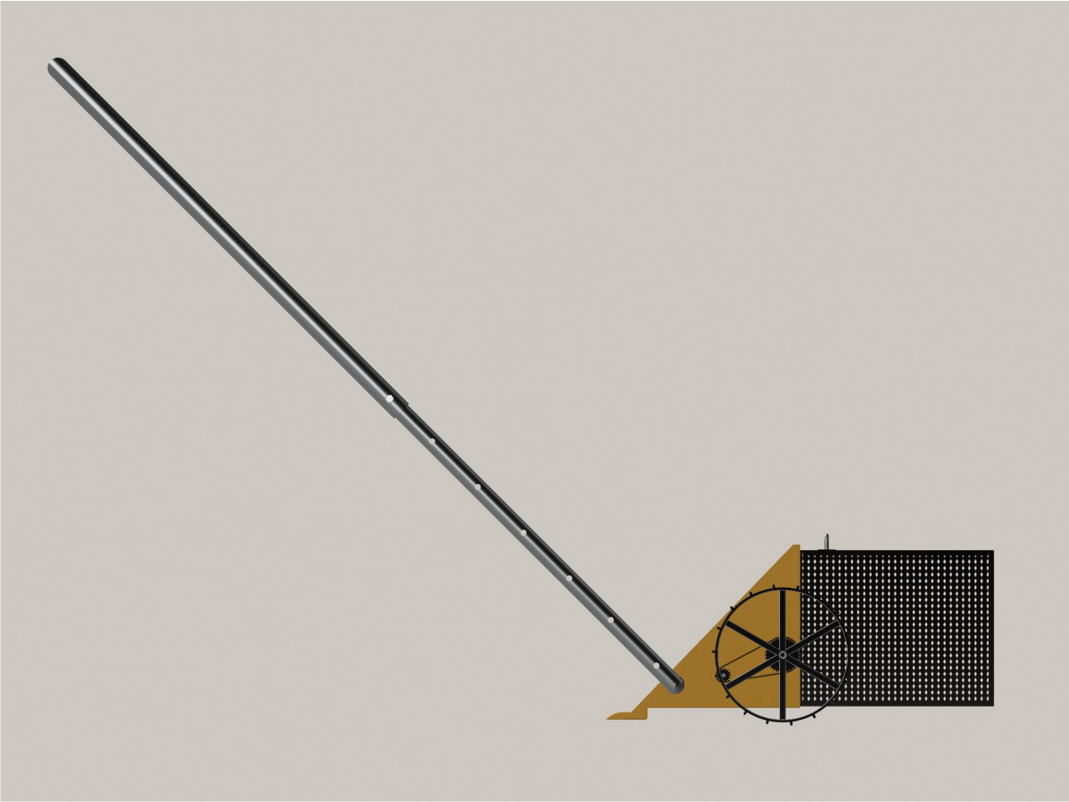

Thank You